Unit 3 - Session 7

SLO 1: GHT for Ellipses & Object Location- Develop dimension reduction strategies for ellipse parameters

Duration: 10 Minutes Activity

Question:

PART A — The Full 5D Ellipse Accumulator

A Generalized Hough Transform for ellipses must search over all five parameters simultaneously. Each parameter is divided into discrete bins. The TOTAL accumulator size equals the product of bins across every dimension. Even a modest bin count explodes in 5D — this is the curse of dimensionality that makes naive GHT impractical.

Param	Description	Range	Resolution	Bin Count
cx	Centre x	0 - 512 px	4 px	128 bins
cy	Centre y	0 - 512 px	4 px	128 bins
a	Semi-major	10 - 200 px	5 px	38 bins
b	Semi-minor	5 - 150 px	5 px	29 bins
θ	Orientation	0 - 180 deg	10 deg	18 bins

STEP 1 — Compute Baseline Total Bins

`Total = cx_bins × cy_bins × a_bins × b_bins × θ_bins`

Total = 128 × 128 × 38 × 29 × 18 = 324,993,024

Your answer: 3.24993024 × 10^8 bins (write in scientific notation)

PART B — Apply Reduction Strategies One-by-One

For each strategy: state the new bin count and compute the running total after applying all strategies so far.

S1 — **Fix Aspect Ratio (b = 0.6 × a)**

b is no longer a free parameter.

New b bins: 1

New Total: 11,206,656 bins

S2 — **Quantize Orientation to 4 bins (0, 45, 90, 135 deg)**

θ resolution coarsened from 10 deg to 45 deg.

New θ bins: **4**
New Total: 2,490,368 bins

S3

Use Edge Gradient to Decouple θ

θ removed from shared accumulator (gradient direction fixes θ per vote).

New θ bins: **1 (removed)**
New Total: 622,592 bins

S4

Reduce Spatial Resolution to 8 px bins

cx and cy bins halved: 512 / 8 = 64 each.

New cx, cy bins: **64 each**
New Total: 155,648 bins

PART C — Reduction Journey Summary

Fill in the computed total bins for each stage. Calculate the percentage remaining relative to baseline (100%).

Baseline (5D)	After S1 (fix b/a)	After S2 (coarse θ)	After S3 (drop θ)	After S4 (coarse cx,cy)
324,993,024 bins	11,206,656 bins	2,490,368 bins	622,592 bins	155,648 bins
100%	3.4483%	0.7663%	0.1916%	0.0479%

[Row 1: Fill in total bin count. Row 2: Percentage of baseline = Stage Total / Baseline Total x 100]

PART D — Cumulative Parameter Bin Count Table

Track all parameter bin counts after each strategy is applied. Fill in the TOTAL BINS column.

Stage	cx	cy	a	b	θ	TOTAL BINS
Baseline	128	128	38	29	18	324,993,024
After S1	128	128	38	1	18	11,206,656
After S2	128	128	38	1	4	2,490,368
After S3	128	128	38	1	1	622,592
After S4	64	64	38	1	1	155,648
Reduction Factor						Baseline / Final = 2,088×

BONUS — Memory Cost Estimate

Each accumulator bin stores a 2-byte integer (uint16)
Baseline memory = Total_bins × 2 bytes = 619.88 MB
Final memory = Total_bins × 2 bytes = 304.00 KB
Memory saved = 99.9521 % (compute reduction ratio)